

Math 4441: Probability and Statistics

Prof. Muhammad Mahbub Alam, PhD

Department of Computer Science and Engineering

Islamic University of Technology

Chapter 2: Random Variables (Part 1)

Lectures 07 – 08

Math 4441: Probability and Statistics

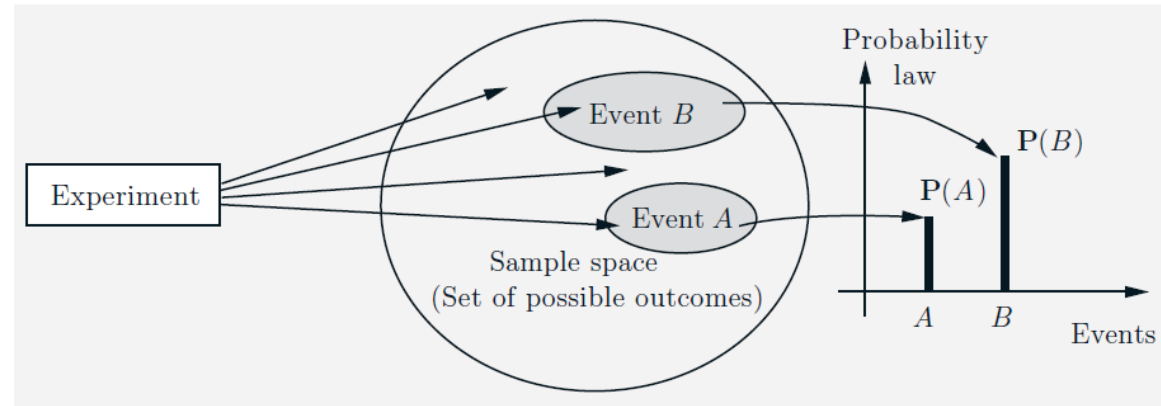
References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Summary of Chapter 1:

Simple Probability Models

1. Sample space (S)
2. Probability of events ($P[\blacksquare]$)



Limitation of the simple probability model:

Sample Space: S

- Elements of S can be anything
- Does not facilitate further processing

Probability of Events: $p[\cdot]$

Lack a concise Representation of probabilities

Example 2.1: (Example 1.12)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

Probabilities of Outcomes	$P[\text{number of success is } 0] = (1 - p)^3$
$P[FFF] = (1 - p)^3$	$P[\text{number of success is } 1] = 3p(1 - p)^2$
$P[FFD] = p(1 - p)^2$	$P[\text{number of success is } 2] = 3p^2(1 - p)$
$P[FDF] = p(1 - p)^2$	$P[\text{number of success is } 3] = p^3$
$P[FDD] = p^2(1 - p)$	
$P[DFF] = p(1 - p)^2$	
$P[DFD] = p^2(1 - p)$	
$P[DDF] = p^2(1 - p)$	
$P[FDD] = p^3$	

$p \triangleq$ probability that a single packet is delivered

Each of the deliveries is independent of the others

What do we need?

- Each element of S is a number
 - Convert each element $\omega \in S$ into a real number $x \in R$.
 - Define a function
 - $X: S \rightarrow R, \quad S_X = \{x_1, x_2, \dots, x_n\}$
 - What does the number x_i represent?
- Probability that the result is x_i
 - Calculate the probability of each $x_i \in S_X, P[X = x_i]$
- Representation of the probability model as Mathematical expression
 - Represent both the real numbers $(x_i, i = 1, \dots, n)$ and their probabilities $P[X = x_i], i = 1, \dots, n)$ as mathematical expressions

Probability Models

Random Variable:

Random variables express the outcome of an experiment by real numbers

- Finds an event space for the experiment
- Define a function that converts the events into real numbers

Distribution Function:

A distribution function represents a collection of probabilities – how probabilities are distributed over the number line

- Each probability is related to a real number, x
- Represents the chance of occurring the event represented by x

Type of Random Variables:

- Discrete random variables
 - Possible values are from a discrete set
 - Number of values are either finite or countably infinite
- Continuous random variables
 - Possible values are from an interval
 - Number of values are uncountable
- Mixed random variables

Example 2.1: (Continued)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

Events related to the observations $E_i \triangleq \# \text{ of success(es) is } i$ $i = 0, 1, \dots, 3$ $E = \{E_0, E_1, E_2, E_3\}$ $S_X = \{ \quad \quad \}$	$X(\omega) = x, \quad X: S \rightarrow R$ FFF FFD FDF DFF FDD DFD DDF DDD 0 1 2 3 S R
--	--

Definition (Random Variable)

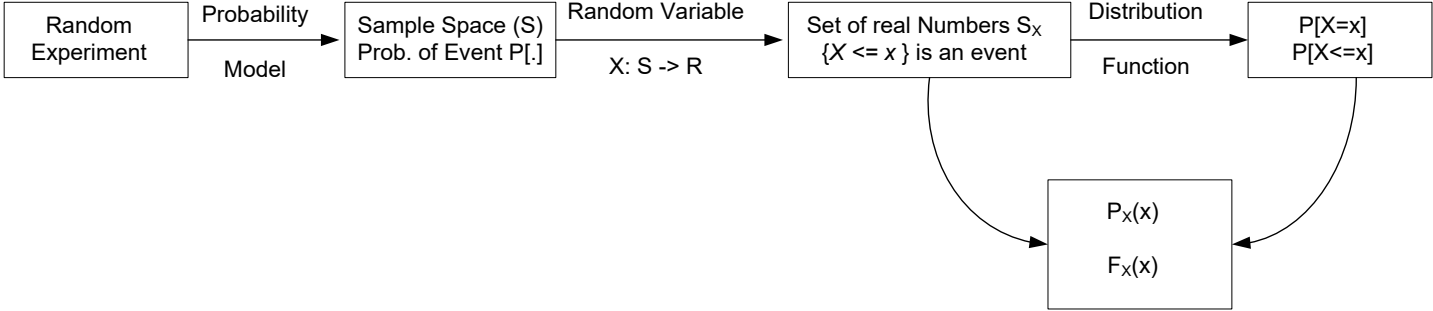
The point function $X(\omega)$ is called a *random variable* if

- (a) Finite real-valued function defined on S ,
- (b) Maps each outcome $\omega \in S$ into a point on the real line
$$X: S \rightarrow R$$
- (c) S_X represents the set of real numbers associated with the random variable
- (d) For every real number x , the set $\{\omega: X(\omega) \leq x\}$ is an event.
- (e) $X(\omega)$ is the function, and we suppress the ω to represent random variables
- (f) We represent a random variable by X

Distributions Functions:

- Represent the distribution of probabilities on the number line
- A probability is attached to each number on the number line
- Probabilities are non-zero for a number, only if the random variable can take on that value
- Represent probabilities associated with the event related to $x \in S_X$

Probability Models by Random Variables



Events associated with a random variable are:

- The random variable has a specific value
 - $\{X = x\}$ or more specifically $\{X(\omega) = x\}$
- The random variable has a value which is less than or equal to a specific value:
 - $\{X \leq x\}$ or $X(\omega) \leq x\}$
- The random variable has value which is greater than a specific value
 - $\{X > x\}$ or $\{X(\omega) > x\}$

Possible distribution functions are:

- Probability Mass function (PMF): The probability that a random variable X has a specific value x
 - $P[X = x]$
- Cumulative distribution function (CDF): The probability that a random variable has a value which is less than or equal to a specific value x
 - $P[X \leq x]$
- Complementary cumulative distribution function (CCDF): The probability that a random variable has a value which is greater than a specific value x
 - $P[X > x]$

Example 2.1: (Continued)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

$$E = \{E_0, E_1, E_2, E_3\}$$

$X \triangleq$ Random variable that counts the number of successes

$$S_X = \{0, 1, 2, 3\}$$

	E_0	E_1	E_2	E_3
S	FFF	$FFD \ FDF \ DFF$	$FDD \ DFD \ DDF$	DDD
x				
$P[X = x]$				
$P[X \leq x]$				

$p \triangleq$ Probability that a single packet is successfully delivered.

Probability Mass Function (PMF): $P_X(x)$

If the set of possible values of X , $S_X = \{x_1, x_2, \dots, x_n\}$, then

1. $P_X(x_i) = 0$, if $x_i \notin S_X$

2. $P_X(x_i) = P[X = x_i]$, $x_i \in S_X$

and $P_X(x_i) > 0$,

3. $\sum_i^n P_X(x_i) = 1$

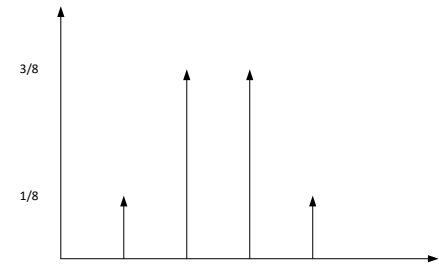
Example 2.1 (Continued)

$$P_X(x) = \begin{cases} \quad, & x = 0 \\ \quad, & x = 1 \\ \quad, & x = 2 \\ \quad, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

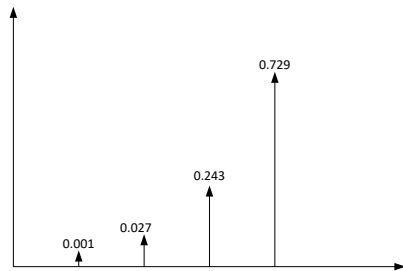
For $p = \frac{1}{2}$,

$$P_X(x) = \begin{cases} \quad, & x = 0 \\ \quad, & x = 1 \\ \quad, & x = 2 \\ \quad, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

Graphical Representation of the PMF



$p = 1/2$



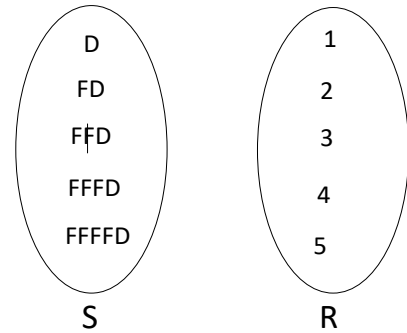
$p = 0.9$

Example 2.2 (Example 1.13 Continued)

Procedure: Keep sending packets from a sender to a receiver until 1 packet is delivered

Observation: Number of attempts

$$S = \{D, FD, FFD, FFFD, \dots\} \quad s_X = \{1, 2, 3, \dots\}$$



	E_1	E_2	E_3	E_4	E_5
S	D	FD	FFD	$FFFD$	$FFFFD$
x					
$P[X = x]$					
$P[X \leq x]$					

Example 2.3: (Example 1.14 Continued)

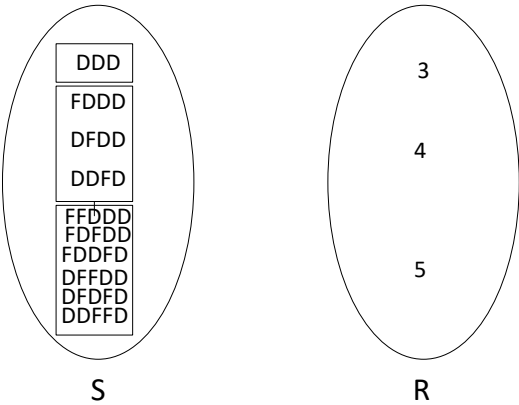
Example 1.14 (**Experiment 1.5**)

Procedure: Keep sending packets from a sender to a receiver until 3 packets are delivered

Observation: Number of attempts

$$S = \{DDD, FDDD, DFDD, DDFD, FFDDD, \dots\}$$

$$S_x = \{3, 4, 5, \dots\}$$



	E_3	E_4	E_5
S	DDD	$FDDD \ DFDD \ DDFD$	$FFDDD \ FDFDD \ FDDFD$ $DFFD,DFDFD,DDFFD$
x			
$P[X = x]$			
$P[X \leq x]$			

Cumulative Distribution Function (CDF)

$$F_X(x) = P[X \leq x]$$

Probability that X will assume a value from the subset of S , where the subset is the point x and all the points to the left of x .

Accordingly, the CDF of X can be defined in terms of its PMF:

$$F_X(x) = \sum_{\substack{y \in S_X, \\ y \leq x}} P_X(y)$$

Example 2.1: (Continued)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

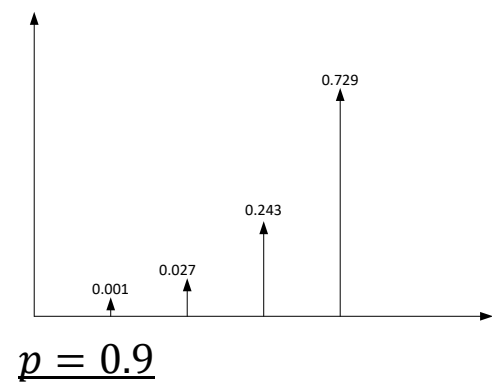
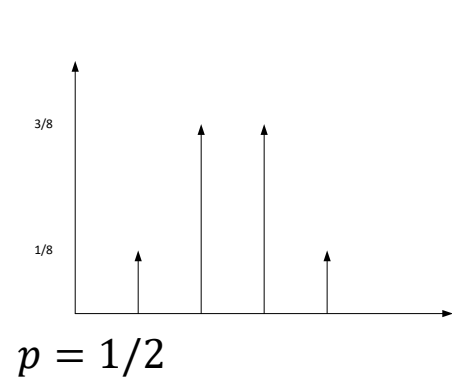
$E = \{E_0, E_1, E_2, E_3\}$ $X \triangleq$ Random variable that counts the number of successes

$$S_X = \{0, 1, 2, 3\}$$

	E_0	E_1	E_2	E_3
S	FFF	$FFD \ FDF \ DFF$	$FDD \ DFD \ DDF$	DDD
x	0	1	2	3
$P[X = x]$	p	$p(1 - p)$	$3p(1 - p)^2$	$3p(1 - p)^3$
	1/8	3/8	3/8	3/8
$P[X \leq x]$				

$p \triangleq$ Probability that a single packet is successfully delivered.

Graphical Representation of the PMF



Example 2.2 (Example 1.13 Continued)

Procedure: Keep sending packets from a sender to a receiver until 1 packet is delivered

Observation: Number of attempts

$S = \{D, FD, FFD, FFFD, \dots\}$ $s_x = \{1, 2, 3, \dots\}$

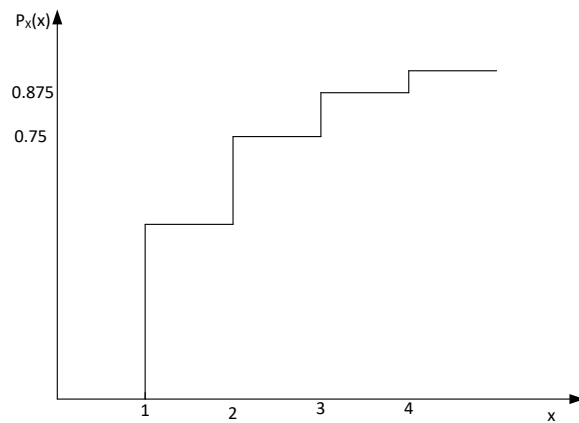
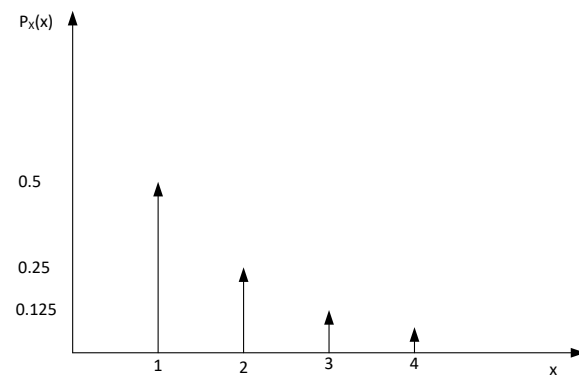
	E_1	E_2	E_3	E_4	E_5
S	D	FD	FFD	$FFFD$	$FFFFD$
x					
$P[X = x]$					
$P[X \leq x]$					

$F_X(x) = 1 - (1 - p)^x, \quad x \geq 1$

Example 2.1 (Continued)

For $p = \frac{1}{2}$,

$$P_X(x) = \begin{cases} 0.5, & x = 1 \\ 0.25, & x = 2 \\ 0.125, & x = 3 \\ 0.0625, & x = 4 \\ 0, & \text{otherwise} \end{cases}$$



Properties of CDF:

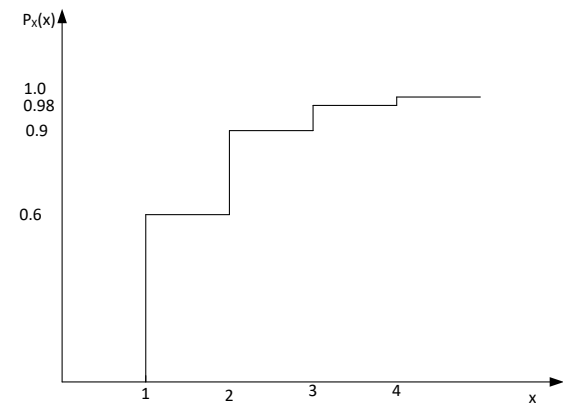
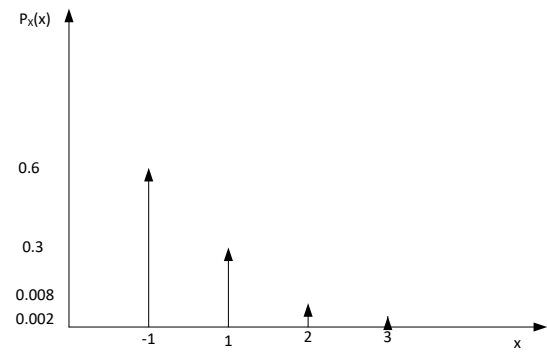
1. It is applicable to both discrete and continuous RVs
2. It is nonnegative, non-decreasing function of x
3. For discrete random variables it is a step function
 - a. Jumps at the values of x where $P_X(x) > 0$
 - b. For continuous random variables, it is continuous
4.
 - a. $F_X(-\infty) = 0$
 - b. $F_X(+\infty) = 1$
5. If a and b are two real numbers such that $a < b$
$$P[a < X \leq b] = F_X(b) - F_X(a)$$

Which is a direct result of

$$P[X \leq b] = P[X \leq a] + P[a < X \leq b]$$

Example 2.4: Let a discrete random variable X assumes values $-1, 1, 2$, and 3 , with probabilities $0.6, 0.3, 0.08$, and 0.02 , respectively. Find the PMF and CDF of X .

$P_X(x) = \begin{cases} 0.6, & x = -1 \\ 0.3, & x = 1 \\ 0.008, & x = 2 \\ 0.002, & x = 3 \\ 0, & \text{otherwise} \end{cases}$	$F_X(x) = \begin{cases} 0, & x \leq -1 \\ 0.6, & -1 \leq x < 1 \\ 0.9, & 1 \leq x < 2 \\ 0.98, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$
---	---

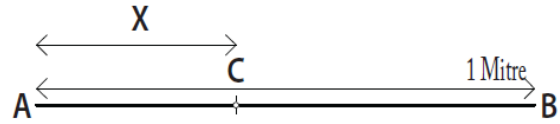


Continuous Random Variables

- Set of possible values of a random variable is uncountable or denumerable
- Set of values are represented by a range of values or by an interval
 - The delay of a packet to reach the destination from the source.
- What is the probability that a continuous random variable has a specific value?

Probability Models of Continuous Random variables

Consider a line AB of length 1 unit.

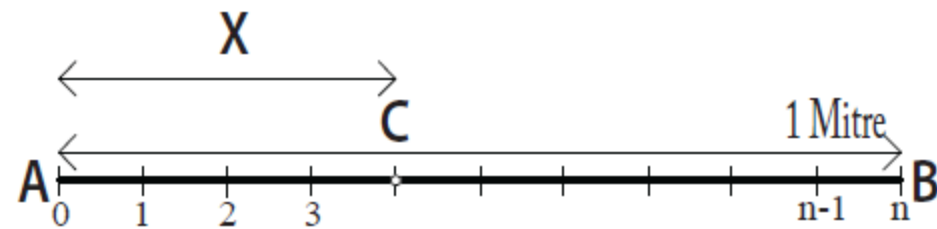


Suppose you randomly choose a point C within AB that divides the line into two parts AC and CB

Let, the length of the point C from A , AC , and the random variable that denote the length of AC is X

Since $S_X = (0, 1)$ is an interval, X is a continuous random variable

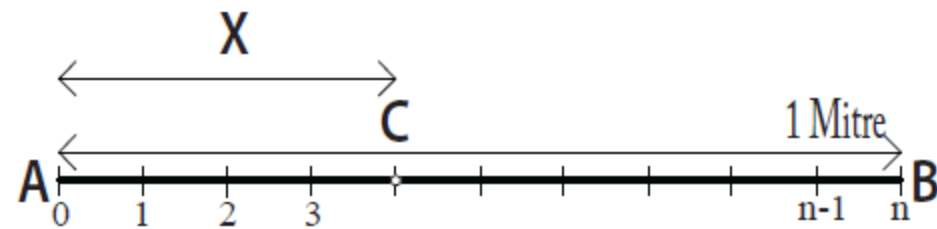
There are infinite possible points in between A and B , therefore, the probability that X has any specific value is $\frac{1}{\infty}$, i.e., it is zero.



To develop a probability model of X , let us consider a reasonable discrete approximation of X

Let us divide the line segment into n equal segments, each numbered from 1 to n

Since all segments are equal in length, if we randomly select a point from AB , it is equally likely that the selected point will be on any specific segment.



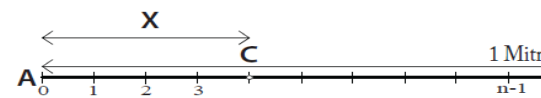
Let Y denote a discrete random variable, representing the number of the segment on which the random point lies.

The range of values of Y , S_Y , is

$$S_Y = \{1, 2, \dots, n\}$$

Two important questions to be answered are:

1. The relation between X and Y
2. How well does Y approximate X .



From the Figure, we can easily see that

$$Y = \lceil nX \rceil$$

If we denote $\{X = x\}$ and $\{Y = \lceil nX \rceil\}$ as two events, we have

$\{X = x\} \subset \{Y = \lceil nX \rceil\}$, and it implies

$$P[X = x] \leq P[Y = \lceil nX \rceil] = \frac{1}{n}$$

$$P[X = x] \leq \lim_{n \rightarrow \infty} P[Y = \lceil nX \rceil] = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$P[X = x] \leq 0$$

According to the 1st Axioms of Probability $P[X = x] \geq 0$, hence $P[X = x] = 0$

PMF as a Probability Model for Continuous Random Variables:

Since $P_X(x) = P[X = x]$ and $P[X = x] = 0$ for continuous random variables,

- The probability that a continuous RV has a specific value is always zero.
- The MPF is meaningless for a continuous random variable.

Distribution Functions for Continuous Random Variables

- Let Random variable X represent the length AC
- Probability of a range of values, an interval, is well defined
 - $P[a < X \leq b]$ is well defined
- All points in $[0,1]$ are equally likely to be selected as point C
- Assume $a = 0$ and $b = 0.5$, intuitively we can say that 50% of the time point C will be selected in the interval $[0, 0.5]$
- Probability that X has a value between 0 and 0.5 is

$$P[a < X \leq b] = P[0 < X \leq 0.5] = \frac{0.5}{1.0} = 0.5$$

- Further assume that $a = -\infty$ and $b = x$ then

$$P[a \leq X \leq b] = P[-\infty < X \leq x] = P[X \leq x] = F_X(x)$$

Cumulative Distribution Function (CDF) of Continuous Random Variables

The cumulative distribution function of a continuous random variable X is denoted as $F_X(x)$, and is defined as

$$F_X(x) = P[X \leq x]$$

And is represented by

$$F_X(x) = \begin{cases} 0, & \text{for } x < 0; \\ x, & \text{for } 0 \leq x < 1; \\ 1, & \text{for } x \geq 1. \end{cases}$$

Properties of CDF for Continuous Random Variables

1. The CDF of a continuous random variable for a value of $-\infty$ is zero:

$$F_X(-\infty) = P[X \leq -\infty] = 0.$$

2. The CDF of a continuous random variable for a value of $+\infty$ is one, i.e.,

$$F_X(+\infty) = P[X \leq +\infty] = 1.$$

3. The probability that a continuous random variable for a value within the interval $(a, b]$ can be given in terms of its CDF

$$P[a < X \leq b] = F_X(b) - F_X(a).$$

4. For continuous random variables, following four possibilities are equal due to the fact that $P[X = x] = 0$.

$$P[a \leq X \leq b] = P[a < X \leq b] = P[a < X < b] = P[a \leq X < b].$$

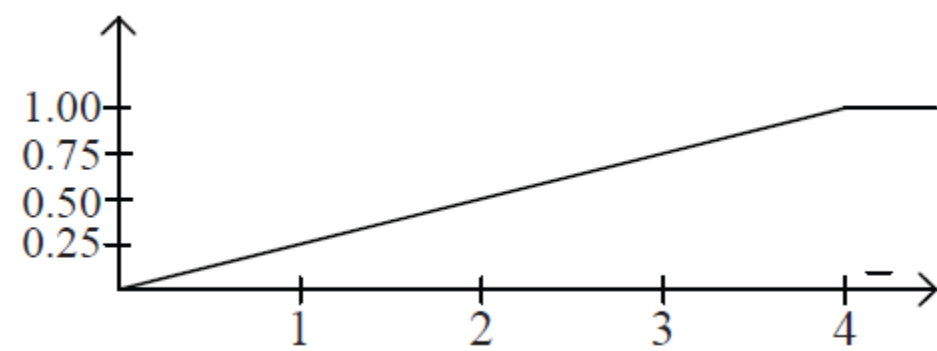
5. For continuous random variables, if $a < b$, then $F_X(a) < F_X(b)$.
6. The CDF of a continuous random variable is non-decreasing and continuous function of x .

Example 2. 10: The CDF of a continuous random variable is

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ \frac{x}{4}, & 0 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

a) Draw the CDF curve.

Find the values of $F_X(-1)$, $F_X(1)$, $P[2 < X \leq 3]$ and $F_X(1.5)$.



$$F_X(-1) = 0.$$

$$F_X(1) = \frac{1}{4}.$$

$$P[2 < X \leq 3] = F_X(3) - F_X(2) = \frac{3}{4} - \frac{2}{4} = \frac{1}{4}.$$

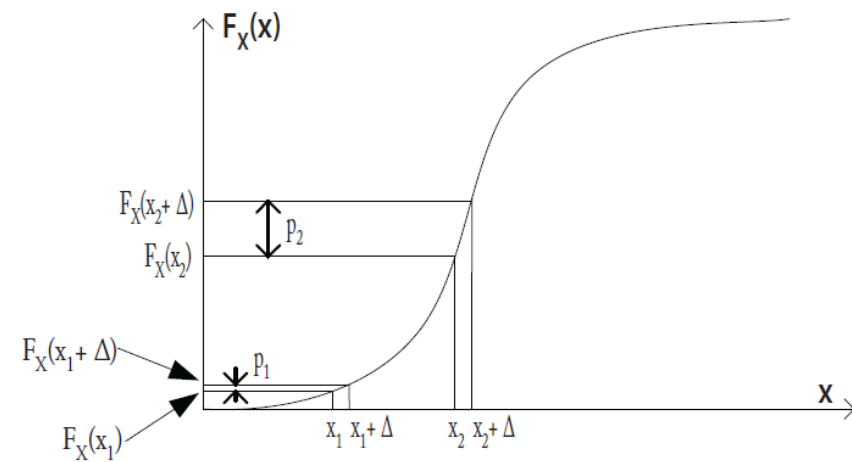
Probability Density Function (PDF)

Distribution of probability for points on the number line

For continuous Random variable defined in $(0, 1)$

- Distribute one unit of probability in the interval $[0, 1]$ on the number line
- Infinite points, cannot assign probability to a specific point, $P[X = x] = 0$
- Though, can assign probability for a range of values, e.g., 0.25 unit in $[0, 0.25]$ 0.50 unit in $[0, 0.50]$
- Distribution can be uniform or non-uniform

We lack to quantify the amount of probability for a specific value of X

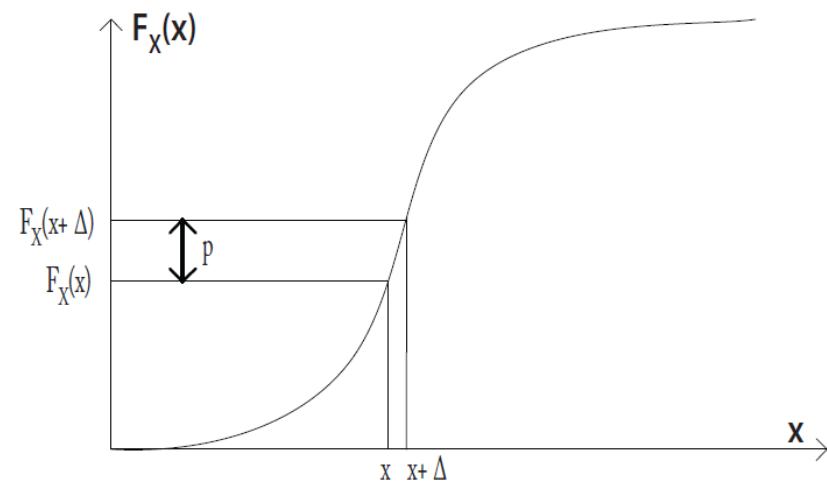


p_1 : probability that X is within x_1 and $x_1 + \Delta$

p_2 : probability that X is within x_2 and $x_1 + \Delta$

$$\begin{aligned} p_1 &= P[x_1 < X \leq x_1 + \Delta] \\ &= F_X(x_1 + \Delta) - F_X(x_1) \end{aligned}$$

$$\begin{aligned} p_2 &= P[x_2 < X \leq x_2 + \Delta] \\ &= F_X(x_2 + \Delta) - F_X(x_2) \end{aligned}$$



Rate of change of probability is higher if the curve is steeper

$$\begin{aligned}
 p &= P[x < X \leq x + \Delta] \\
 &= F_X(x + \Delta) - F_X(x) \\
 &= \frac{F_X(x + \Delta) - F_X(x)}{\Delta} \times \Delta
 \end{aligned}$$

Average amount of Probability per unit length

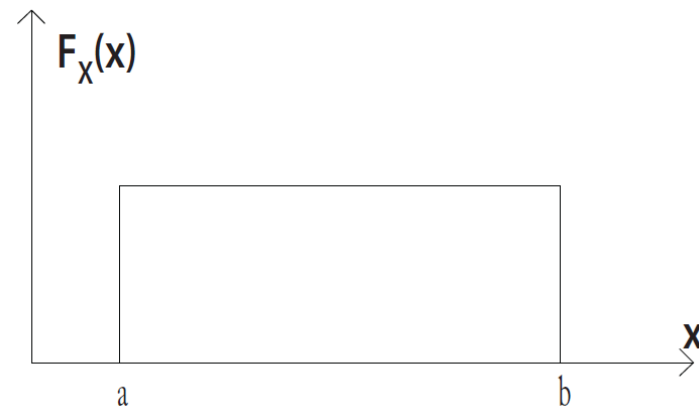
Density: measure of the amount of mass per unit space (usually volume)

Probability Density: Measure of the amount of probability per unit length

$$f_X(x) = \lim_{\Delta \rightarrow 0} \frac{F_X(x + \Delta) - F_X(x)}{\Delta}$$

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Area Under the Curve (AUC)



$$P[a < X \leq b] = \int_a^b f_X(x) dx.$$

$$\int_{-\infty}^{+\infty} f_X(x) dx = 1$$

$$\begin{aligned}P[-\infty < X \leq x] &= \int_{-\infty}^x f_X(x)dx \\&= P[X \leq x] \\&= F_X(x).\end{aligned}$$



Example 2.11: For the CDF $F_X(x)$ given below

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ \frac{x}{4}, & 0 \leq x < 4, \\ 1, & x \geq 4 \end{cases}$$

find the following:

1. $f_X(x)$, from the CDF
2. $F_X(x)$, from the PDF
3. $P[2 < X \leq 3] = ?$
4. Draw the PDF curve

$$\begin{aligned}
 f_X(x) &= \frac{d}{dx} F_X(x) \\
 &= \frac{d}{dx} \frac{x}{4} \\
 &= \frac{1}{4}.
 \end{aligned}$$

$$f_X(x) = \begin{cases} \frac{1}{4}, & \text{for } 0 \leq x \leq 4; \\ 0, & \text{otherwise.} \end{cases}$$

$$\begin{aligned}
 F_X(x) &= \int_{-\infty}^x f_X(x)dx \\
 &= \int_0^x \frac{1}{4}dx \\
 &= \left. \frac{x}{4} \right|_0^x \\
 &= \frac{x}{4} - \frac{0}{4} = \frac{x}{4}
 \end{aligned}$$

$$F_X(x) = \begin{cases} 0, & \text{for } x < 0; \\ \frac{x}{4}, & \text{for } 0 \leq x < 4; \\ 1, & \text{for } x \geq 4. \end{cases}$$

$$\begin{aligned}P[2 < X \leq 3] &= \int_2^3 f_X(x)dx \\&= \int_2^3 \frac{1}{4}dx \\&= \frac{x}{4} \bigg|_{x=2}^{x=3} \\&= \frac{3}{4} - \frac{2}{4} = \frac{1}{4}\end{aligned}$$

Example 2.13: The PDF of a continuous random variable X is

$$f_X(x) = \begin{cases} cxe^{-x/2}, & \text{for } x \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the value of the constant c .
2. Find the CDF of the random variable X .
3. Find the probability $P[2 \leq X \leq 5]$ from the PDF of X .
4. Find the probability $P[2 \leq X \leq 5]$ from the CDF of X .

$$\begin{aligned}
 \int_{-\infty}^{+\infty} f_X(x)dx &= \int_0^{\infty} cxe^{-x/2}dx \\
 &= cx(-2)e^{-x/2} \Big|_{x=0}^{x=\infty} - \int_0^{\infty} 1.(-2)e^{-x/2}dx \\
 &= 0 + 2c \int_0^{\infty} e^{-x/2}dx \\
 &= 2c(-2)e^{-x/2} \Big|_{x=0}^{x=\infty} \\
 &= 0 + 4c = 4c
 \end{aligned}$$

$$4c = 1 \text{ and } c = \frac{1}{4}.$$

$$\begin{aligned}
F_X(x) &= \int_{-\infty}^x \frac{1}{4} x e^{-x/2} dx \\
&= \frac{1}{4} x (-2) e^{-x/2} \Big|_{x=0}^{x=\infty} - \frac{1}{4} \int_0^{\infty} 1 \cdot (-2) e^{-x/2} dx \\
&= -\frac{1}{2} x e^{-x/2} + \frac{1}{2} \int_0^{\infty} e^{-x/2} dx \\
&= -\frac{1}{2} x e^{-x/2} + \frac{1}{2} (-2) e^{-x/2} \Big|_{x=0}^{x=\infty} \\
&= -\frac{1}{2} x e^{-x/2} - e^{-x/2} + 1 \\
&= 1 - e^{-x/2} - \frac{1}{2} x e^{-x/2}.
\end{aligned}$$